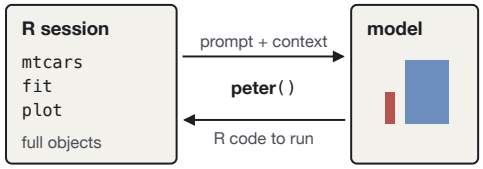
Agents in your R session with gptr :: CHEATSHEET

`peter()` runs language-model agents on the objects already in your R session. Named for Peter Wason, whose dual-process research inspires its System 1 and System 2 roles, and Peter Naur, of Backus-Naur form: sessions become readable, runnable documents.

How it works



Objects stay in R; the agent's code runs there on the full data. An agent is sent the prompt, bounded previews, a workspace listing, the R code you ran since the last prompt, project instructions and tool output (`?gptr_egress`; `.opts = list(context = "none")` drops listing and code); a decision model, each input.

gptr_describe(`mtcars`)
What a model is shown of an object.

Set up

pak::pak(`"Broccolito/gptr@main"`)
library(`gptr`)
Install gptr 1.0 from GitHub (R 4.2 or later).

gptr_env(`"~/keys/.env"`)
Load keys from a private .env file; gptr redacts them.

gptr_login(`"openrouter"`)
Sign in interactively: OAuth for OpenRouter, masked key entry for other providers.

gptr_providers()
Each route's credential, status, default model, egress.

gptr_models(`"sonnet"`)
Search the offline catalog by name, alias or provider.

gptr_config(`model = opus, .scope = "user"`)
Save a default model. Scopes: session, project, user.

gptr_init(`"."`)
Create a `.gptr/` workspace so sessions and recorded answers outlive R (else they end with it); at the console it then asks whether to trust the project.

gptr_trust(`".", trust = TRUE`)
Trust without the question, as in a script, or change it.

gptr_permissions(`allow = "r(fn:saveRDS)"`)
Add a rule for this R session; no arguments lists rules.

`fake = `**gptr_fake_provider**(`list("32 rows.")`)
A scripted offline model for examples and tests.

No model set? Interactive `peter()` (no prompt) offers the Codex CLI, the Claude Code CLI or manual API or Ollama setup, and saves a CLI choice as default. `peter("...")` takes the first ready route instead (an Anthropic, OpenAI or Google key, else Claude Code, else Codex) and asks once before sending context.

Ask Peter from R

`peter("...")` returns a session, in scripts, loops and functions too. Where nobody can approve (Rscript, knitr), pass `mode = auto` or an allow rule, or the first change stops the run (`gptr_error_permission`).

`s = `**peter**`("Fit mpg on wt; save it as fit.", mtcars)`
The prompt first, then the objects to work on.

s$text; s$value; s$turns
Last answer, newest `gptr_return()`, prompt count.

s$usage; s$cost; s$envir
Usage, cost (NA without prices) and workspace (NULL in a function unless `envir = new.env()`).

`col = "mpg"; `**peter**`("Plot {col} by cyl.")`
Literal prompts interpolate `{name}` from R.

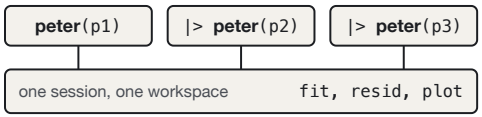
peter`("Clean the data.", raw, mode = auto, model = "openai/gpt-6-sol")`
Name other arguments exactly; bare names work.

peter`("Summarise.", sales, budget = list(tokens = 5e4, turns = 5))`
Per-call limits, checked before each request; `turns` counts model requests, `cost` needs priced models.

gptr_return`(coef(fit))  # in agent R code`
Marks the result, `s$value` until a later one replaces it.

tryCatch`(`**peter**`("Fit.", d), gptr_error = function(e) e$session)`
A stopped run carries `e$session` (after Esc or Ctrl-C, `gptr_last()`); `gptr_rewind(e$session)` drops its failed turn. A call refused up front changes nothing.

Pipe: one session, many turns



`s |> `**peter**`("Now plot the residuals.")`
Same session and workspace, one more turn.

peter`("Draft a summary.", fit) |> `**peter**`("Shorten it.")`
Each pipe adds a turn to the session on its left.

`b = `**gptr_fork**`(s)`
`b |> `**peter**`("Try a log model instead.")`
Branch off; only in-place edits (data.table `:=`, environments, `<<-`) reach the original's objects.

`s |> `**peter**`("Add a caveat.", model = opus)`
Switch the model mid-pipe; the history is kept.

`s; `**summary**`(s); `**gptr_last**`()`
Print it, its history, the most recent session.

The console

`session = `**peter**`()`
`session |> `**peter**`()`
Open the console; `/exit` returns the session.

AT THE peter> PROMPT

`Fit mpg on wt.`	A prompt for the session
`!head(d)`	Run R and show the model
`!!x = 2`	Run R; only object changes are sent
```` ```r … ``` ````	Fenced R, run like `!code`
`""" … """`	A multi-line prompt
`line \`	Continue on the next line
`@notes.md`	Attach a file's first 40 lines
`@fit`	Attach an object
`{n}`	Insert an R value

SLASH COMMANDS

`/help [cmd]`	Commands and syntax
`/model [ref]`	Show or switch model
`/mode [mode]`	Show or switch mode
`/status`	Session, model, mode
`/cost /context`	Tokens and cost
`/env [text]`	Workspace objects
`/tools`	Tools and helpers
`/compact [focus]`	Compact the history
`/clear`	New chat, same objects
`/resume [id]`	List or resume sessions (across R restarts only in a `.gptr/` project)
`/fork`	Fork and continue there
`/doc [path]`	The history document
`/skills /skill:name`	List or use skills
`/mcp`	List MCP servers
`/permissions`	Show or change rules
`/undo /redo /retry`	Undo, redo, resend
`/rewind [k]`	Keep turns 1 to k
`/exit`	Leave; return the session (NULL if none yet)

WHILE AN ANSWER RUNS

In a terminal (not Windows), type, then Enter steers and Tab follows up. Ctrl-C (RStudio: Esc) pauses: **[s]**teer, **[f]**ollow-up, **[c]**ontinue, **[a]**bort, **[b]**ackground (in the console it runs on only after `/exit` or `!!gptr_wait(gptr_last())`). Approvals (same key aborts): **[y]**es, **[a]**lways (session), **[p]** always (project), **[n]**o (`n <text>` says what to do instead), **[?]**.

System 1 and System 2

System 1 — typed decision → `TRUE "lung" 1.88` vector + probabilities

System 2 — agent with tools → `gptr_session` $text, $value, objects

The model decides what `peter()` returns. A decision model (Jev, Clef) answers once per input; a conversational model works in turns.

SYSTEM 1: TYPED DECISIONS

`rct = `**peter**`("Is this an RCT?", abstracts, model = jev)`
A logical vector: TRUE when P(yes) ≥ `threshold` (0.5).

gptr_prob`(rct)`
P(yes) per element; `"probabilities"` gives the matrix.

peter`("Which tissue?", cells, model = jev, choices = c("liver", "lung", "other"))`
One label per input: a character vector.

peter`("How severe?", events, model = jev, levels = c("low", "mid", "high"))`
A score: the expected zero-based level, such as 1.88.

peter`(q, x, model = jev, min_confidence = 0.8, uncertain = "stop")`
Abstain below a confidence: for yes/no |2p−1| (0.8 keeps p ≤ 0.1 or ≥ 0.9), for choices and scores the engine's own scale. NA by default; `TRUE/FALSE` (yes/no) or `function(state, answer)` fills in; `"stop"` errors, naming the inputs.

`abstracts[rct]; `**ifelse**`(rct, "RCT", "other")`
Plain vectors for `if`, `for`, subsetting and `ifelse()`.

`s |> `**peter**`("Did it succeed?", model = jev)`
Judge a session's last answer; adds no turn.

`m = if (isTRUE(hard)) "opus" else "haiku"`
peter`(task, model = m)`
Route each task to a deep or a fast agent.

SYSTEM 2: AGENTS

peter`("Plan it.", d, model = opus, .opts = list(thinking = "high"))`
More reasoning: `off` to `max`, or a model suffix such as `:high`; a model without that level uses its nearest one and says so (opus has no `off`).

Try it offline

```
step = list(tool = "r", input = list(
  code = "fit = lm(mpg ~ wt, mtcars)"))
fake = gptr_fake_provider(list(step, "Done."))
s = peter("Fit mpg on wt.", mtcars,
  model = fake, mode = auto); coef(fit)
```

No key: the first reply runs R, the second is `s$text`.

Models and providers

```
"anthropic/claude-sonnet-5-5:high"
```
provider · model id · thinking

Aliases work too: `sonnet`, `opus`, `haiku`, `gpt`, `flash`, `jev`, `claude_code`, `codex`.

`anthropic/…`	Anthropic `ANTHROPIC_API_KEY`
`openai/…`	OpenAI `OPENAI_API_KEY`
`google/…`	Google Gemini `GEMINI_API_KEY`
`openrouter/…`	OpenRouter; also groq, deepseek, mistral, together, xai, cerebras and fireworks routes, listed after `gptr_models(refresh = TRUE)` <NAME>_API_KEY
`azure/…`	Azure OpenAI `AZURE_OPENAI_API_KEY`, `AZURE_OPENAI_ENDPOINT`
`bedrock/…`	AWS Bedrock `AWS_BEARER_TOKEN_BEDROCK`, `AWS_REGION`
`ollama/…`	Ollama, local, no key. Also lmstudio, llamacpp, vllm: tools once a model is declared with `tool_call = TRUE`
`claude-cli`	Claude Code CLI `claude auth login`
`codex`	Codex CLI `codex login`
`jev`	System 1 · TypeSafe Jev `TYPESAFE_API_KEY`
`ollama/clef`	System 1 · local Clef and Clef Flash, Ollama 0.35.1 or later

gptr_models(provider = "ollama",
  refresh = TRUE)
Ask Ollama which models are installed; for a cloud route, download the models.dev catalog of all routes.

```
m = list(id = "my-id", tool_call = TRUE)
gptr_config(providers = list(
  bedrock = list(models = list(m))))
```
Declare Bedrock or Azure models, then `model = "bedrock/my-id"`. Each call replaces `providers`.

gptr_register(**gptr_provider**("corp",
  "openai-completions", base_url = u,
  auth = "CORP_KEY", models = list(m)))
Any OpenAI-compatible endpoint; registering the name again replaces it. `api = "bedrock-converse"` with `compat = list(region = "us-west-2")` signs Bedrock with SigV4: `auth` names the secret key, `AWS_ACCESS_KEY_ID` holds the key id.

```
eg = gptr_config()$egress
eg$openai = "ack"
gptr_config(egress = eg, .scope = "user")
```
Acknowledge egress to openai, for unattended runs.

The script is the history

Bind the document with your `peter()` calls; each top-level call gets a block of the agent's recorded R code.

```
peter("Fit mpg on wt in mtcars.")
# >>> gptr:46d369 model=…
fit = lm(mpg ~ wt, mtcars)
coef(fit)[["wt"]]
#> [1] -5.344472
## Decision: linear fit
# <<< gptr:46d369
```
prompt · header · code · output · note · end

gptr_doc("analysis.R")
Record this R session's calls into the `.R`, `.Rmd`, `.qmd` or `.ipynb` holding them; `quarto render` and Knit start another R session, so bind in the setup chunk.

gptr_config(mode = auto, .scope = "session")
Under `Rscript` or `rmarkdown::render()` nobody can approve: record with auto mode (or the call's `mode =`) or `gptr_permissions()` rules; replays need none.

gptr_source("analysis.R",
  replay = "replay")
Rerun: fresh blocks replay with no model call; a new R session gets `s$text` back only from `.gptr/`.

```
GPTR_REPLAY=replay Rscript analysis.R
```
Prove from the shell that a script needs no model.

gptr_blocks("analysis.R")
Each block's status: fresh, stale (prompt edited), user-edited or undone.

gptr_doc("report.Rmd")
rmarkdown::render("report.Rmd")
Render as usual; blocks become `gptr-<id>` chunks.

gptr_doc("analysis.ipynb", sync = TRUE)
After closing the notebook, run this, in any R session, to write the blocks queued while it was open.

gptr_doc("log.R") # at the console
Append each console turn to a runnable `.R` transcript as `s_<id> = peter("…")`, with `!code` lines and slash commands. R typed at the > prompt is not recorded, so create inputs with `!code`.

peter(q, replay = "live")
Modes: `"auto"` (default), `"replay"`, `"live"`, `"record"`. `"live"` re-asks the model under `gptr_source()`, knitr, an IDE or Jupyter; plain `source()` and `Rscript` replay a recorded call instead, with a warning.

EXPORT A RECORDED .R

Export needs fresh blocks and `peter()` calls whose arguments are all quoted strings, numbers or TRUE/FALSE (write `mode = "auto"`): no data objects, `.opts` or {name} prompts; `s |> peter()` only continues an `s` an earlier `peter()` statement assigned.

gptr_export("analysis.R", "plain.R")
Plain R: recorded code runs, `peter()` calls become comments; later code reading `s` (`s$value`) is refused.

gptr_export("analysis.R", "peter.R",
  format = "peter")
`peter()` calls run; recorded code is commented.

Tools, files and artifacts

The model calls `read`, `r`, `edit`, `write` and `ask`; scripts offer `ask` only in `manual` mode, where it stops the run. Inside `r` it uses `peter$` helpers, which you can call too.

peter("Review R/.", tools = extended)
Presets: `minimal` (no ask), `standard`, `readonly` (read and r only; r can still change files, so use `mode = plan` to forbid changes), `extended` (adds grep, find, ls).

peter(q, tools = c("-write", "+grep"))
Remove or add single tools (a bare name adds; an unknown name is an error).

peter("What is in data/scores.csv?")
Reads text and images; binary files get a loading hint.

peter$read("R/fit.R", limit = 40)
peter$grep("lm\\(", path = "R")
peter$find**("*.R"); **peter$ls("R")
Read a file window, search contents, find and list files.

peter$edit("R/fit.R", list(list(
  oldText = "n = 10", newText = "n = 20")))
peter$write("notes.txt", "done\n")
Exact text, else ignoring trailing spaces and quote, dash and space variants (diff printed). Atomic writes.

peter$sh(c("git", "status"))
peter$bg(c("Rscript", "long.R"))
Run a program; run one in the background.

peter$sql("SELECT cyl, AVG(mpg) FROM mtcars
  GROUP BY cyl ORDER BY cyl", mtcars)
SQL on data frames by name (duckdb). `peter$py()` runs Python once you set `RETICULATE_PYTHON` or call `reticulate::py_config()`.

ARTIFACTS: SHINY APPS

peter("Make a Shiny app of it.", survey)
Writes `app.R` in `.gptr/artifacts/<id>/` (temporary before `gptr_init()`) and, at the console, launches it.

gptr_artifacts()
gptr_artifacts("survey-view", open = TRUE)
List apps by id; open, stop or relaunch a version.

Permissions and trust

`plan`	Read only; R runs in a scratch environment. Its plan goes once to your next `peter()` call (option `gptr.plan_handoff`)
`manual`	Default: read-only actions run, others ask; with nobody to ask, the run stops
`edits`	Also the write and edit tools in the project; R code that writes files (`write.csv()`) still asks
`auto`	All but critical actions, gptr settings and secret reads
`yolo`	Nearly all; critical actions still ask

gptr_permissions(deny = "r(sh:rm *)")
Rules: `deny` beats `ask` beats `allow`.

gptr_rewind(s)
Undo the last turn: objects, files and conversation.

The gate is advisory, not a sandbox: `?gptr_security`.

Skills, plugins and MCP

gptr_skills()
Skills from the project, user, plugins and packages.

peter("Speed up f.", f,
  skills = "high-performance-r")
Preload a skill. Write yours as `<name>/SKILL.md` with a `description:` line in `tools::R_user_dir("gptr", "config")/skills`, or in `.gptr/skills/` of a trusted project.

```
.gptr/prompts/check.md  # becomes /check
```
A console command; project templates need trust.

peter("Report units.", plugins = "./lab")
Enable a plugin; one inside the project loads once the project is trusted. `gptr_plugins()` lists them.

gptr_mcp_add("docs",
  url = "http://127.0.0.1:8000/mcp")
Add an MCP server (or `command =` for stdio).

petermcp<server>$<tool>(...)
MCP tools are R functions the agent calls.

```
h = gptr_mcp_serve(); h$config  # paste one
```
h$stop()
Serve this R session to Claude Code, Codex or Cursor. Calls needing approval are refused: set the mode or rules first.

```
up = gptr_tool("shout", "Upper-case text",
  fun = function(x) toupper(x))
off = gptr_register(up)
```
Add a tool; re-registering replaces it, `off()` removes it.

Teams, fan-outs, background

Members inherit the mode. A team or fan-out raises an error only when no member finishes; otherwise `$status` and `$reason` name the members that did not.

```
s = peter("Review fit.", agents = list(
  stats = agent(system = "Check stats."),
  code = agent(system = "Check code.")))
```
A team of named roles; `s$stats$text` reads one.

```
f = peter("Summarise; gptr_return() it.",
  cohorts, parallel = 2)
```
One child per element, two at a time; `f$text` has each answer, `f$value` each `gptr_return()` value.

gptr_parallel(a = **peter**("Plan."),
  b = **peter**("Review."))
Run independent calls concurrently.

```
b = peter("Check.", fit, background = TRUE)
gptr_wait(b)
```
Experimental: return at once; also `gptr_steer()`, `gptr_cancel()`, `gptr_jobs()`.

gptr_usage(s, by = "agent")
Tokens and cost by session, agent, model or route.

Learn more

Guides: broccolito.github.io/gptr/articles (start with Getting started). `browseVignettes("gptr")` lists them only if the install built vignettes (CRAN, `R CMD build`).